

Technical Lead – Full Stack & AI-Enabled Systems

Company Overview

We are launching a new technology-driven division within a well-established company with over 30 years of industry presence.

This new division will operate as a **startup-style engineering team** focusing on building modern, scalable web and mobile-enabled software solutions using contemporary technologies. While backed by a stable and reputable organization, the division operates with startup-level agility and innovation.

This is a greenfield initiative. The selected candidate will play a foundational role in defining architecture, setting up the engineering environment from scratch, and building the core product.

Role Summary

We are seeking a highly experienced (7-12 years), hands-on Technical Lead to lead the technical execution of this new division.

This is a deeply hands-on technical role with significant individual coding responsibility. This is not a managerial or coordination-only position.

The selected candidate will:

- Own system architecture
- Set up the complete development and deployment environment from scratch
- Lead and mentor developers
- Define engineering standards and DevOps protocols
- Coordinate web and mobile integration
- Integrate AI/ML capabilities developed by separate teams
- Collaborate across backend technologies
- Contribute directly to coding and architecture decisions

This role requires deep technical expertise, strong execution capability, and the ability to build both systems and engineering foundations from zero.

The selected candidate is expected to contribute directly to core modules and remain actively involved in development throughout the lifecycle of the product.

This position is suited for senior engineers who enjoy building systems and writing production-grade code, and not for candidates seeking purely managerial or oversight roles.

Key Responsibilities

1. Architecture & Technical Ownership

- Define scalable, modular system architecture
- Design secure RESTful APIs
- Ensure backend is extensible for AI/ML integrations
- Architect services to support multi-platform consumption (web + mobile)
- Oversee database architecture and optimization
- Define standards for cross-service communication

2. Environment Setup & Engineering Foundations (From Scratch)

- Set up the complete development environment for the team
- Define repository structure and Git branching strategy
- Establish CI/CD pipelines
- Configure cloud infrastructure (Azure or AWS)
- Implement Docker-based deployments
- Define Dev / Staging / Production environments
- Set up logging, monitoring, and security standards
- Establish coding standards and engineering best practices

3. AI/ML Integration & Productionization

- Integrate AI/ML models and services built by data science teams
- Design service layers to consume ML outputs
- Coordinate containerized deployment of models
- Ensure monitoring, scalability, and reliability of AI services
- Manage API versioning and lifecycle management for ML services
- Translate business AI requirements into production-ready workflows

4. Cross-Technology Backend Coordination

- Lead backend architecture primarily in .NET
- Coordinate with backend developers working in Node.js and Spring Boot (Java)
- Define API contracts and communication protocols across services
- Ensure interoperability between different backend stacks
- Establish best practices for microservices or modular architecture

5. Development Leadership

- Lead and mentor development team
- Conduct code and architecture reviews
- Establish coding standards

- Define Git workflows and branching strategies
- Manage technical debt
- Promote clean, scalable, maintainable code practices

The Technical Lead is expected to lead by example through direct code contribution and architectural implementation, not through delegation alone.

6. DevOps & Infrastructure

- Design and oversee CI/CD pipelines
- Manage cloud environments (Azure or AWS)
- Oversee containerization (Docker)
- Ensure proper logging and monitoring systems
- Maintain environment separation (Dev / Staging / Production)

7. Web & Mobile Integration

- Design APIs suitable for both Angular frontend and mobile apps
- Coordinate with mobile development team
- Ensure clean API documentation (Swagger/OpenAPI)
- Align release cycles across platforms

Required Technical Skills

Backend (Core Requirement)

- Strong expertise in .NET Core / .NET 6+
- REST API development
- Authentication (JWT, OAuth)
- SQL database design
- Scalable architecture design

The candidate must have deep hands-on expertise in backend architecture and development using .NET Core.

Frontend

- Strong Angular experience
- Component-based architecture
- State management principles

DevOps & Environment Setup

- CI/CD pipelines
- Docker
- Cloud deployment (Azure or AWS)
- Git workflows
- Experience setting up environments and infrastructure from scratch

AI/ML Integration Knowledge

- Experience integrating ML models via APIs
- Understanding of model serving concepts
- Familiarity with containerized deployments
- Experience integrating external AI services is a plus

Cross-Stack Knowledge (Optional)

- Working knowledge of Node.js OR
- Working knowledge of Spring Boot (Java)
- Understanding of microservices communication patterns
- Ability to coordinate across polyglot backend environments

Required Experience

- 7–12 years of software development experience
- 3+ years in a technical leadership role while remaining actively involved in hands-on coding
- Experience architecting production systems
- Experience setting up engineering environments from scratch
- Experience integrating AI or external services
- Experience leading multi-technology teams is a plus

Must be currently hands-on in development (candidates who have moved into full-time management roles will not be suitable)

Ideal Candidate Profile

The ideal candidate:

- Is deeply hands-on
- Has strong architecture ownership experience
- Has built systems and environments from the ground up
- Has integrated AI/ML features into products
- Has experience coordinating across multiple backend technologies
- Has led engineering teams
- Understands cloud-native application design
- Can operate with startup agility within structured organization

Other Requirements:

This role is foundational to the new division and requires continuity in architecture and technical leadership. The company expects a minimum service commitment of 24 months to ensure stability and successful execution.

The agreed notice period shall be 60 days from either side.

Salary:

22 – 35 LPA + Bonuses (Based on experience and interview rounds)